



WMSU

Introduction to Enterprise Systems for Management

ENTERPRISE SYSTEMS IN ORGANIZATIONS

- ✓ **Enterprise systems (ES)** is a large-scale application software packages that support business processes, information flows, reporting, and data analytics in complex organizations. While ES are generally packaged enterprise application software (PEAS) systems they can also be bespoke, custom developed systems created to support a specific organization's needs.

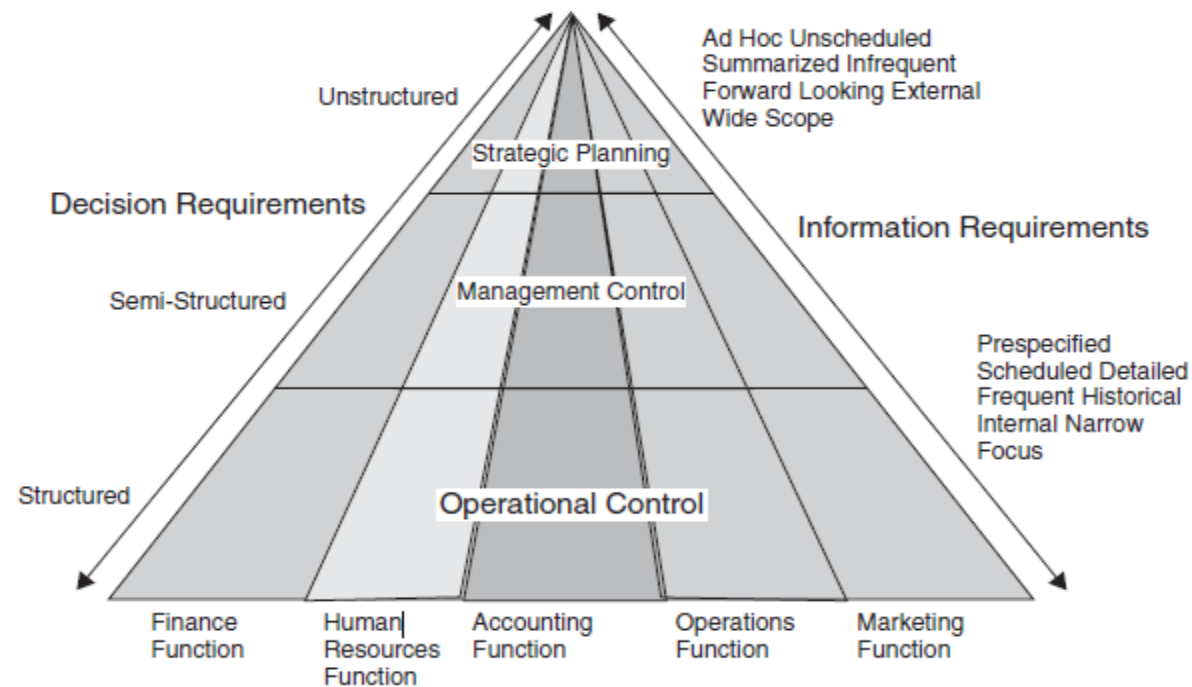


FIGURE 1-1 Management Pyramid with Information Requirements

ENTERPRISE RESOURCE PLANNING SYSTEMS

✓ What Is An ERP?

- **Enterprise resource planning (ERP) systems** are the specific kind of enterprise systems to integrate data across and be comprehensive in supporting all the major functions of the organization.

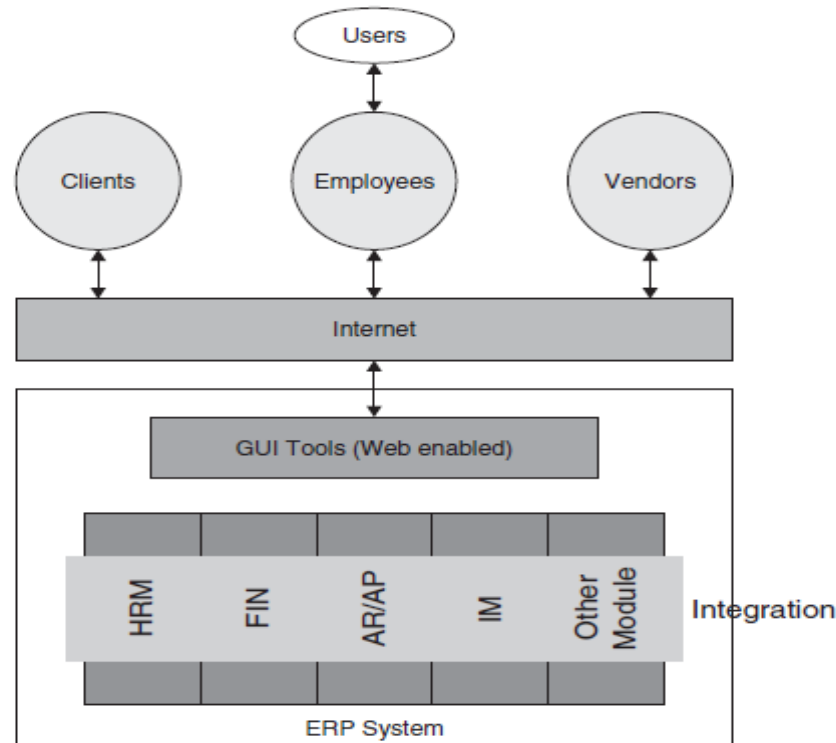


FIGURE 1-2 Integrated Systems—ERP

ERP systems are comprehensive software applications that support critical organizational functions

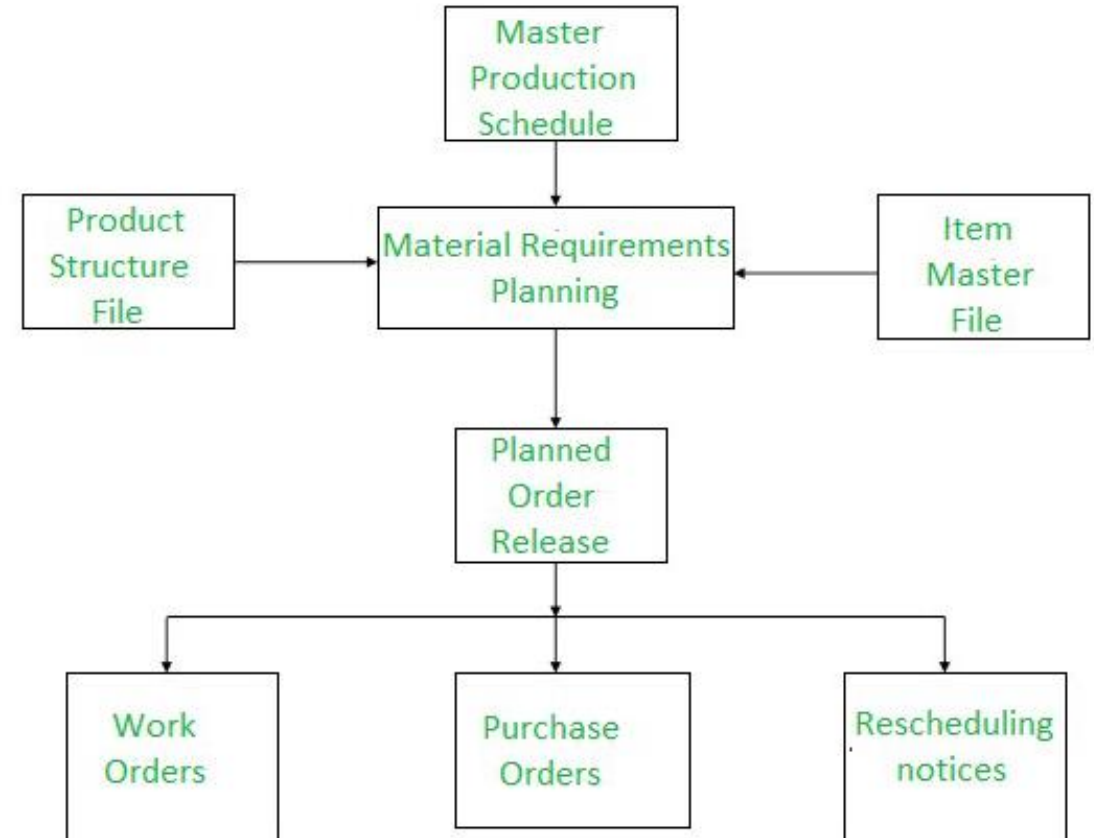
The goal of the ERP :

- is to make information flow be both dynamic and immediate.
- to integrate departments and functions across an organization onto a single infrastructure that serves the needs of each department.

Evolution of ERP System

1. Material Requirement Planning (MRP)

Developed in 1970s, Material Requirement Planning is widely used approach for production planning and scheduling in industry. It is the approach embedded in many commercially available software applications. The function of MRP is to provide material availability, it is used to produce requirement quantities on time. This process involves monitoring of stocks and demand, leading to automatic creation of procurement proposals for purchasing or production. The main objective of MRP is to determine which material is required, quantity required and by when it is required.



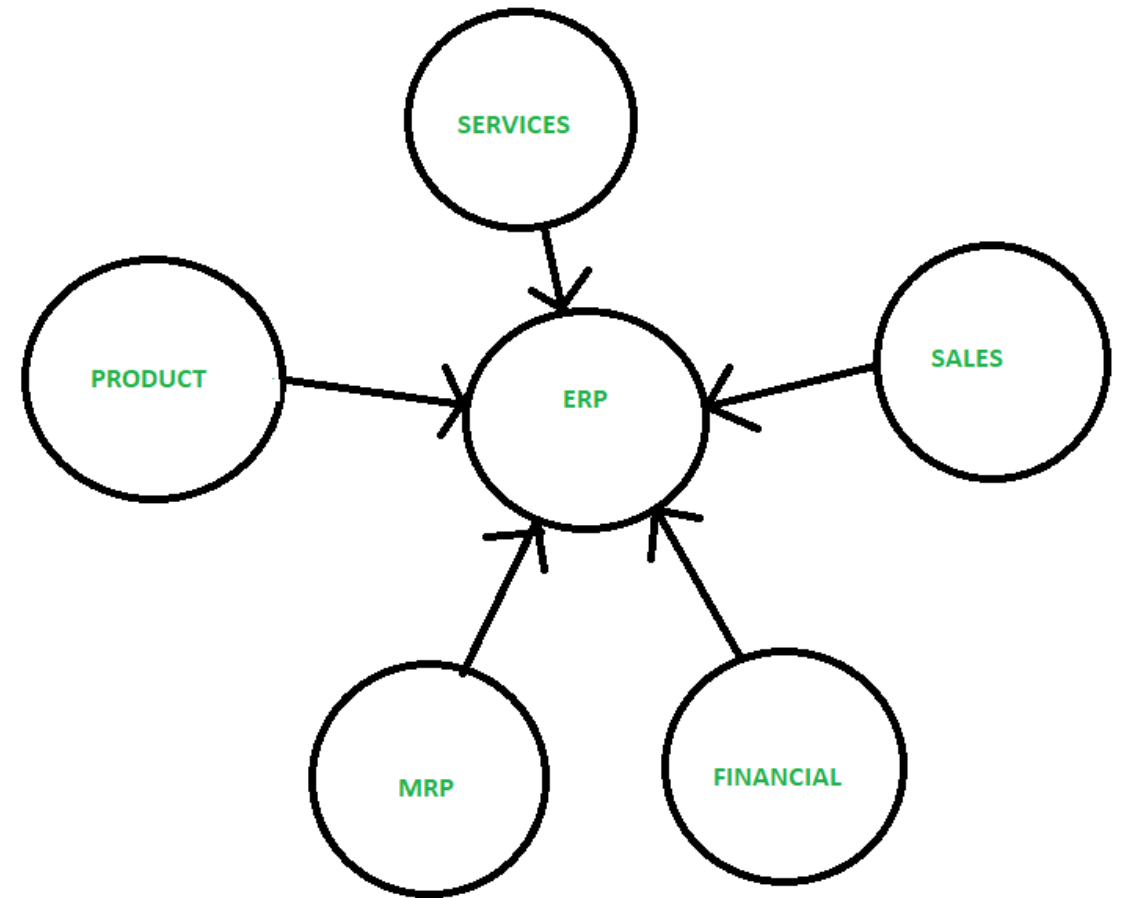
2. Manufacturing Resource Planning (MRP II)

Developed in 1980s, Manufacturing Resource Planning is an expansion of closed loop MRP for managing an entire manufacturing company. This system provides an information that is useful to all functional areas and encourages cross-functional interactions. It supports sales and marketing by providing and orders promising capability. It is a broad-based resource co-ordination system involving other areas of a firm in planning processes, such as marketing, finance and HR.



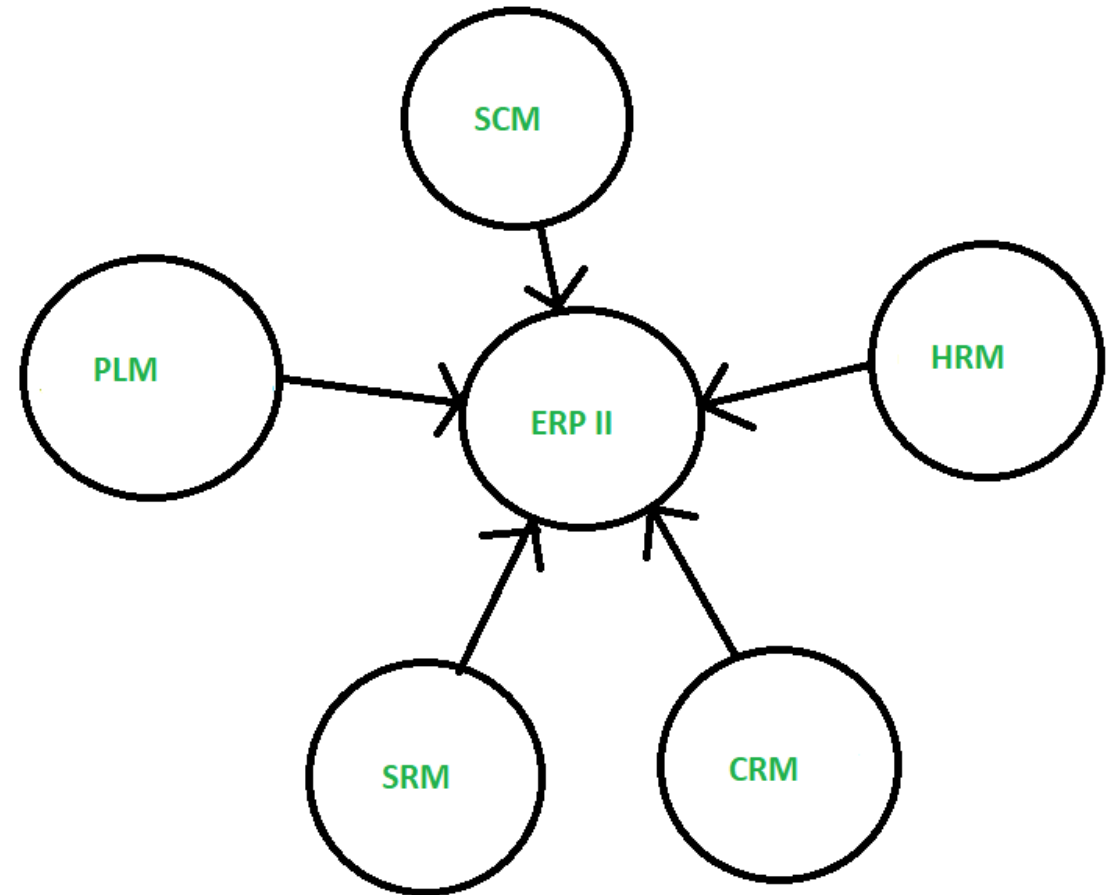
3. Enterprise Resource Planning (ERP)

Developed in 1990s, Enterprise Resource Planning is foundation system for domestic and global operations, supporting most or all functional areas in their daily operations. is one of more common categories of business software, especially with large-scale businesses. It is a business strategy and a set of industry-domain-specific applications that build customer and shareholder communities value network system by enabling and optimizing enterprise and inter-enterprise collaborative operational and financial processes. ERP at its core is an effective way of centralizing information and workflow processes through data management. Because ERP keeps all of your workflow data in one place.



4. Enterprise Resource Planning(ERP II)

Developed in 2000s, ERP II is name now use to describe ERP. Basically, it is successor of ERP. It is a business strategy and set of collaborative operational and financial processes internally and beyond enterprise. These new business models reflect an increased business focus on internal integration. It's domain is in all sectors and segments. Data in this is internally and externally published and subscribed. It includes departmental modules, CRM, SCM and other stakeholders modules. It emphasis on intangible assists.





What is Business Process?

- A business process is a series of tasks or activities grouped to achieve a business function or goal.

Example :

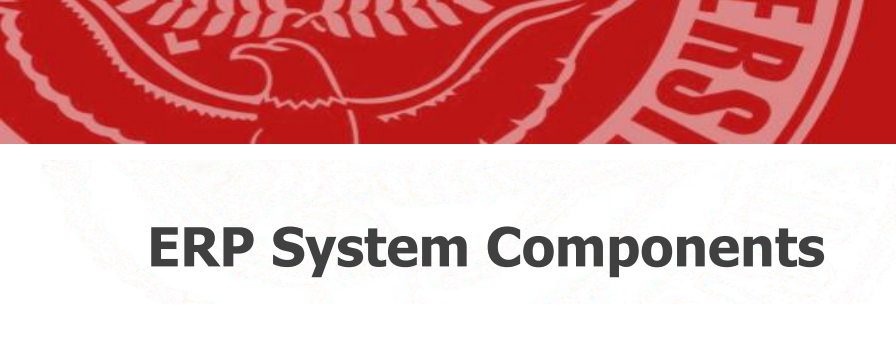
Sales Process

- ✓ sharing the sales proposal
- ✓ sending quotes
- ✓ negotiations
- ✓ receiving orders for product/service
- ✓ updating records of sales
- ✓ delivery of product/service
- ✓ billing
- ✓ Payment

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- The ERP software has hundreds of business processes built into the logic of the system. These processes may or may not agree with the organization's current business processes.

An organization has two choices when implementing ERP:

1. Change business processes to match the software's functionality.
2. Modify the ERP software.



ERP System Components

- ✓ ERP system, like its information system counterpart, has similar components such as hardware, software, database, information, process, and people. These components work together to achieve an organization's goal of enhanced efficiency and effectiveness in their business processes.
- ✓ An ERP system depends on hardware (i.e., servers and peripherals), software (i.e., operating systems and database), information (i.e., organizational data from internal and external resources), process (i.e., business processes, procedures, and policies), and people (i.e., end users and IT staff) to perform the input, process, and output phases of a system. The basic goal of ERP, like any other information system, is to serve the organization by converting data into useful information for all the organizational stakeholders.
- ✓ The key components for an ERP implementation are hardware, software, database, processes, and people.

FIGURE 1-3 ERP Components

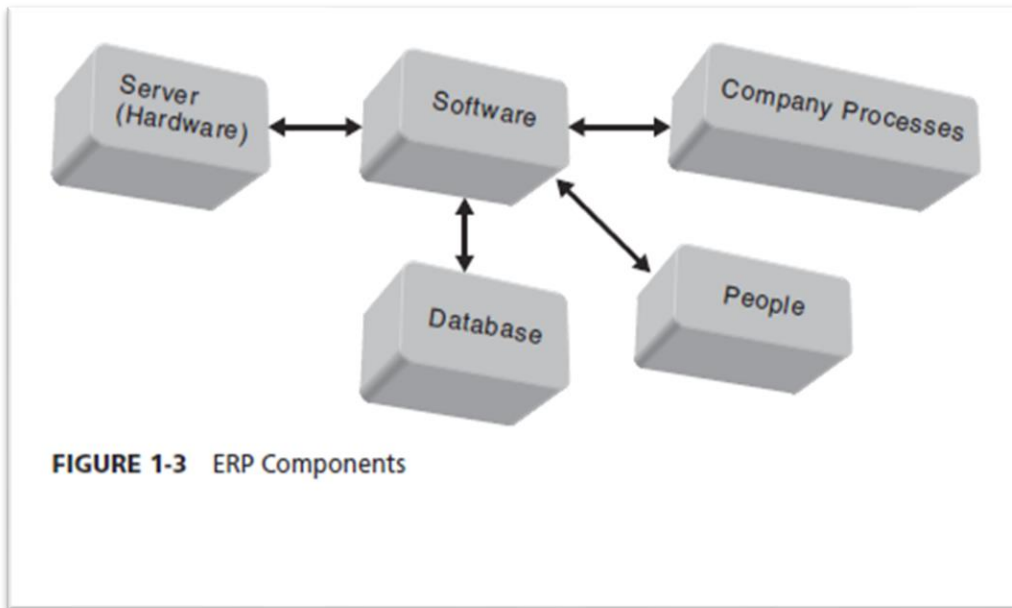
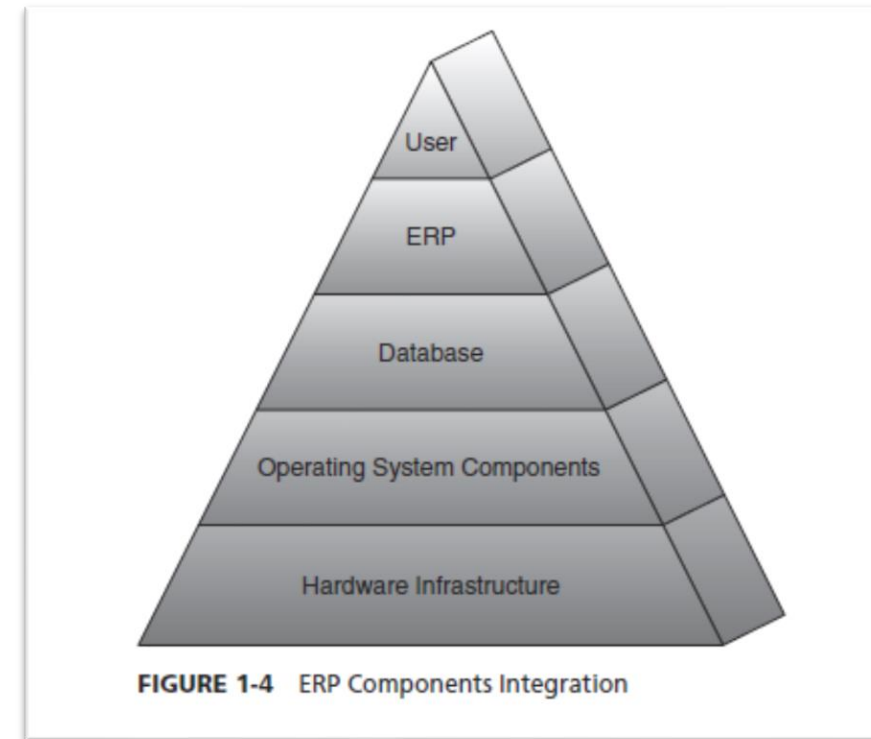


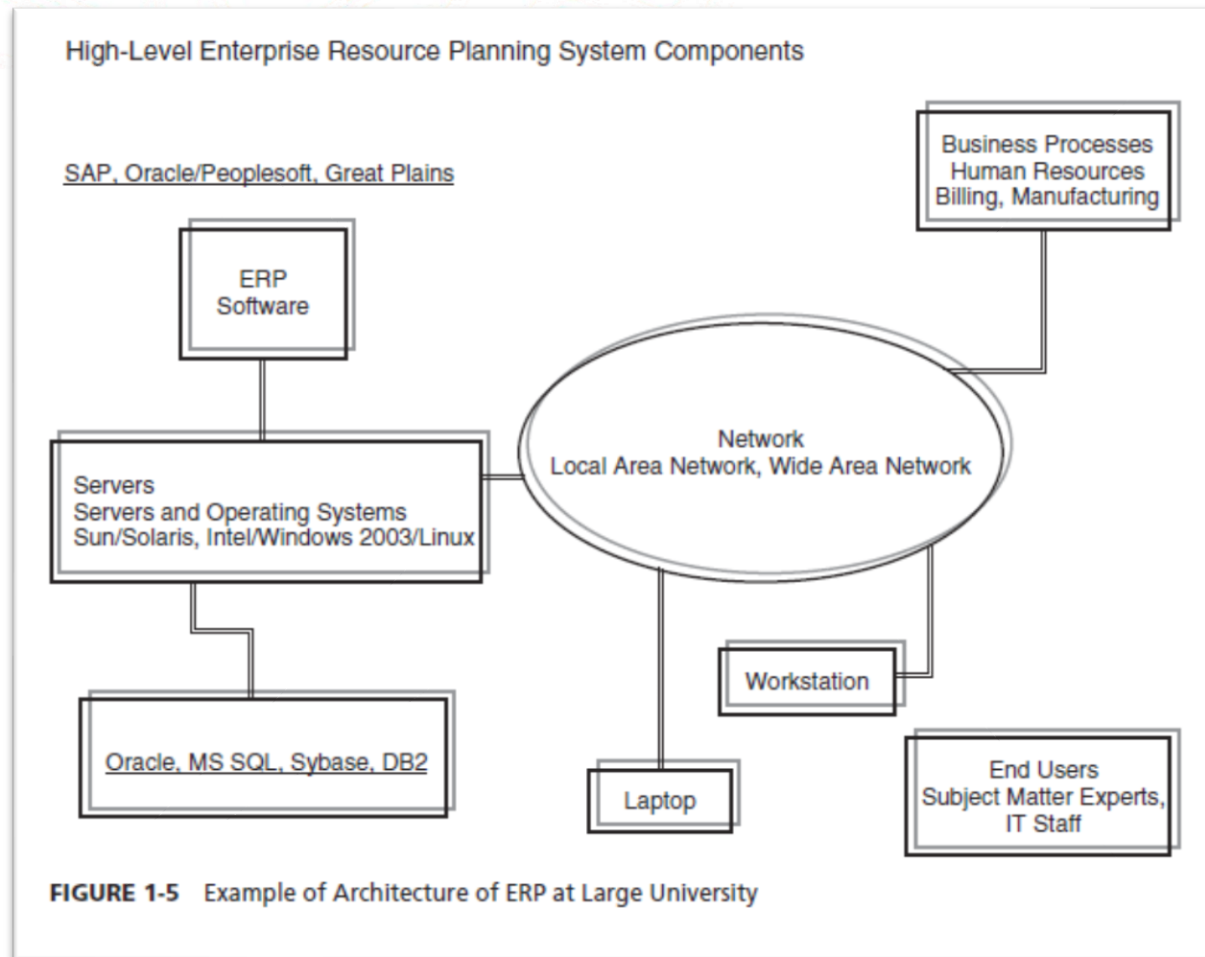
FIGURE 1-4 ERP Components Integration





ERP Architecture

- ✓ The architecture of the ERP implementation influences the cost, maintenance, and the use of the system.
- ✓ A system's architecture is a blueprint of the actual ERP system and transforms the high-level ERP implementation strategy into an information flow with interrelationships in the organization.
- ✓ The ERP architecture helps the implementation team build the ERP system for an organization.
- ✓ An ERP package can have a very different implementation outcome from one organization to another.
- ✓ The architecture sets the stage for modifications or customizations to support an organization's policies and procedures, data conversion, system maintenance, upgrades, backups, security, access, and controls.



In the architecture of a large university, an ERP system can be very complex and must be designed and tested thoroughly before implementing it in the organization (Figure 1-5).

FIGURE 1-5 Example of Architecture of ERP at Large University

Two types of Architecture for ERP system.

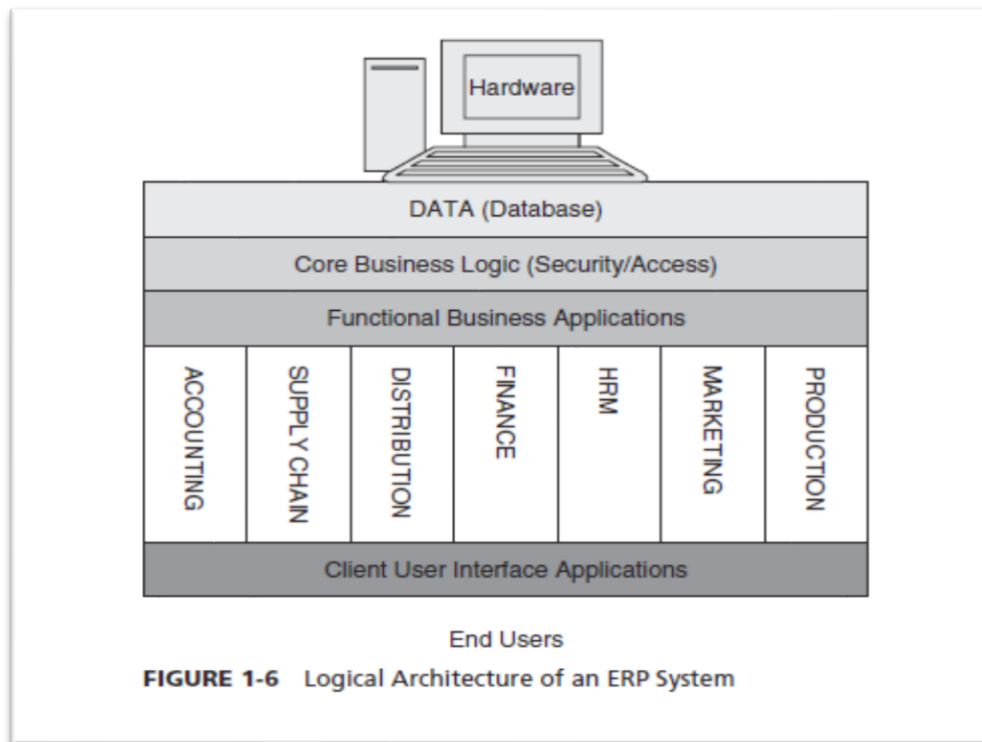


FIGURE 1-6 Logical Architecture

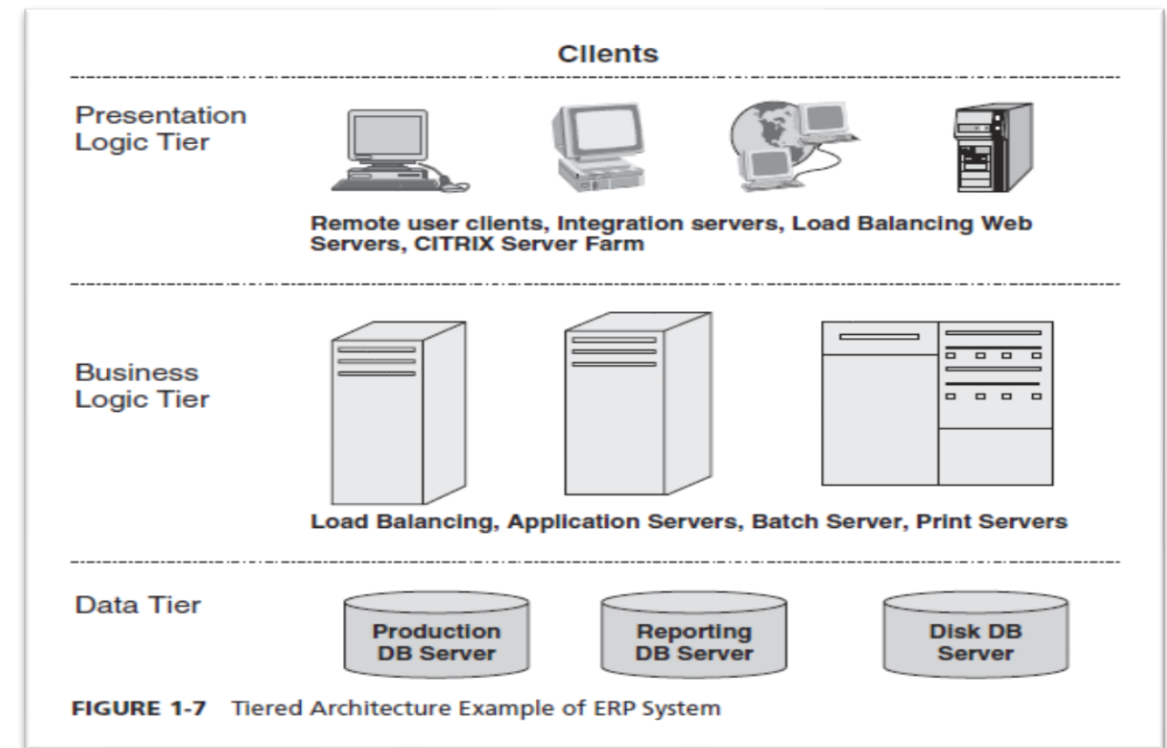


FIGURE 1-7 Physical/Tiered Architecture

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- ✓ The **logical architecture** provides the database schemas of entities and relationships at the lowest tier, followed by the core business processes and business logic handled by the system at the second tier.
 - ✓ The third **tier architecture** provides details on the applications that support the various business functions built in to the ERP system. The end users do not ever see the first and second tiers because they interact primarily with the client–user interface application tier that provides them access to the functional applications.



e-Business and ERP

1. **e-Business technology** focus has been on linking a company with its external partners and stakeholders, whereas **ERP** focus has been on integrating the functional silos of an organization into an enterprise application. *e-Business* technologies that have emerged as successful over the decade (e.g., business-to-consumer and business-to-business) have generally focused on market growth by selling products and services to new consumers and markets. On the other hand, ERP technology has been successful in integrating business processes across the functional spectrum of the organization and in providing a central repository of all corporate data, information, and knowledge, thereby increasing organizational efficiency and worker productivity.
2. **e-Business** is a disruptive technology, whereas **ERP** is adaptive technology. *e-Business* practically transformed the way business operates in terms of buying and selling, customer service, and its relationships with suppliers. This caused a lot of disruptions in organizational strategy, structure, power, and the like. ERP has emerged as an adapter by merging the early data processing and integration efforts within a large corporation.

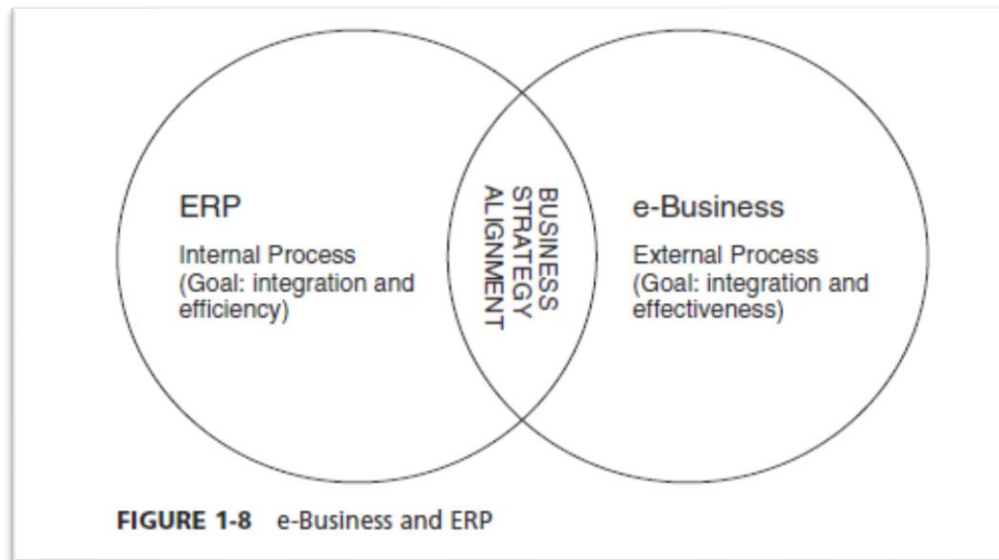


FIGURE 1-8 e-business and ERP

e-Business focus on communication (e-mail), collaboration (calendaring, scheduling, group support), marketing and promotion (Web sites), and electronic commerce.

ERP systems was mainly focus on data sharing, systems integration, business process change, and improving decision making through the access of data from a single source.



Benefits and Limitations of ERP

The system benefits and limitations of ERP systems are follows.

1. Integration of data and applications across functional areas of the organization (i.e., data can be entered once and used by all applications in the organization, improving accuracy and quality of the data).
2. Maintenance and support of the system improves as the IT staff is centralized and is trained to support the needs of users across the organization.
3. Consistency of the user interface across various applications means less employee training, better productivity, and cross-functional job movements.
4. Security of data and applications is enhanced due to better controls and centralization of hardware, software, and network facilities.
5. Complexity of installing, configuring, and maintaining the system increases, thereby requiring specialized IT staff, hardware, network, and software resources.
6. Consolidation of IT hardware, software, and people resources can be cumbersome and difficult to attain.
7. Data conversion and transformation from an old system to a new system can be an extremely tedious and complex process.
8. Retraining of IT staff and personnel to the new ERP system can produce resistance and reduce productivity over a period of time.

The business benefits and limitations of ERP systems are follows.

1. Increasing agility of the organization in terms of responding to changes in the environment for growth and maintaining the market share in the industry.
2. Sharing of information across the functional departments means employees can collaborate easily with each other and work in teams.
3. Linking and exchanging information in real time with its supply chain partners can improve efficiency and lower costs of products and services.
4. Quality of customer service is better and quicker as information flows both up and down the organization hierarchy and across all business units.
5. Efficiency of business processes are enhanced due to business process reengineering of organization functions.
6. Retraining of all employees with the new system can be costly and time consuming.
7. Change of business roles and department boundaries can create upheaval and resistance to the new system.



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